RETAIL AI

Promotion Planner

Tool-assisted promotion planning with inventory, competitor pricing, margin, and budget constraints.

ET AI HACKATHON : Agentic Edition| 2026

Team: Ganesh Vutti



The retail promotion challenge

Manual promotion planning, excess inventory, competitor pricing, and margin constraints.

Promotion decisions must balance clearance, competitiveness, and profitability.

Excess inventory

High stock can tie up working capital and require targeted promotion.

Competitive pricing

Discounts must respond to competitor prices without eroding margin.

Business constraints

Plans must fit a marketing budget and preserve a minimum margin.

The goal

Turn product and pricing data into a transparent, budget-aware promotion proposal.

Our solution

A local AI assistant combines model reasoning with deterministic business tools.

1 Inspect

Read inventory, selling price, cost, and competitor price from a CSV.

2 Calculate

Python computes safe discounts, promotional prices, and post-discount margins.

3 Recommend

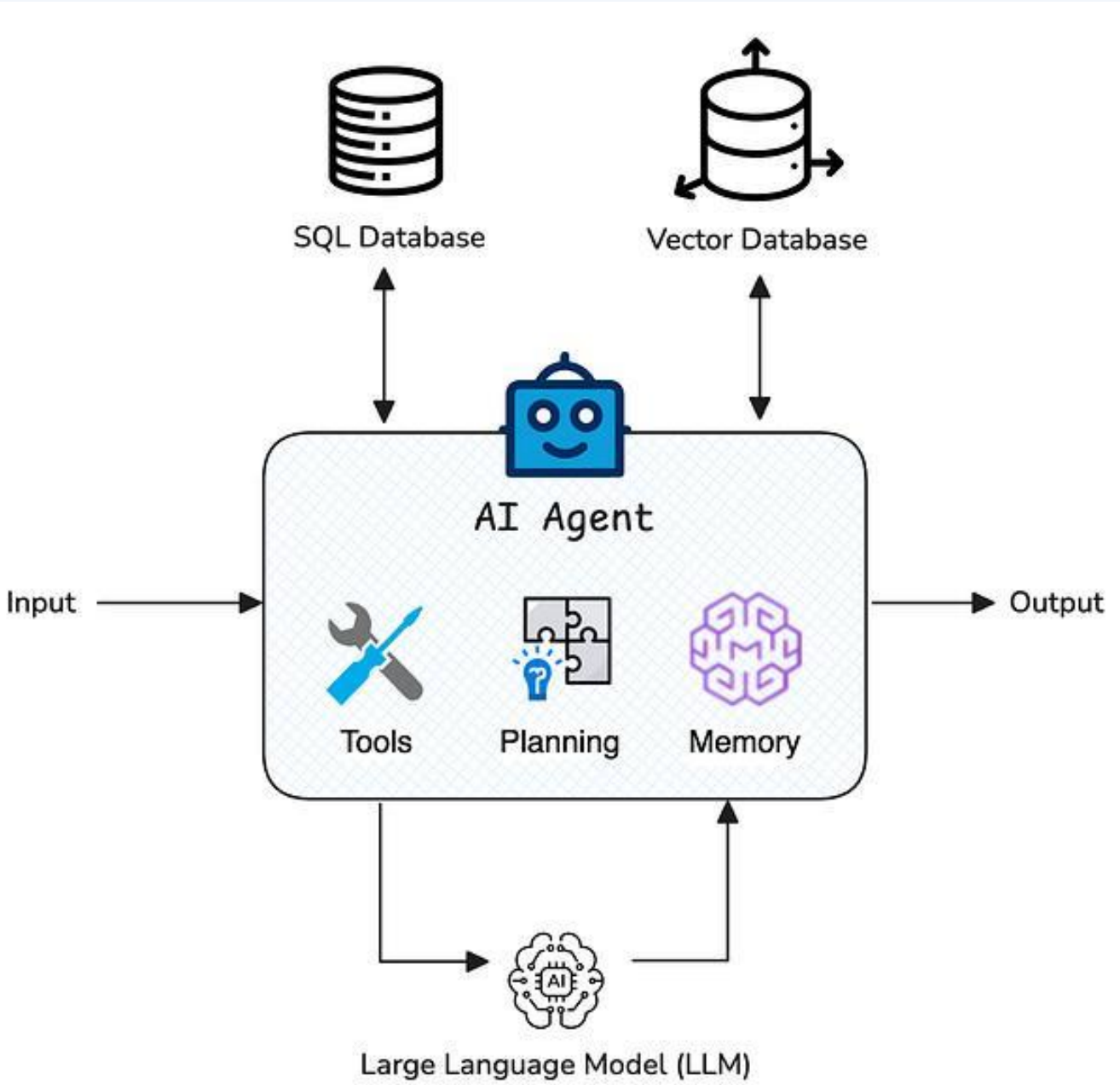
The agent calls tools to build a budgeted plan and explain its choices.

Human-readable outputs

- Streamlit dashboard with inventory and promotion views
- Upcoming weekend opportunities plus manually configured 2026 festival dates
- Downloadable CSV promotion plan

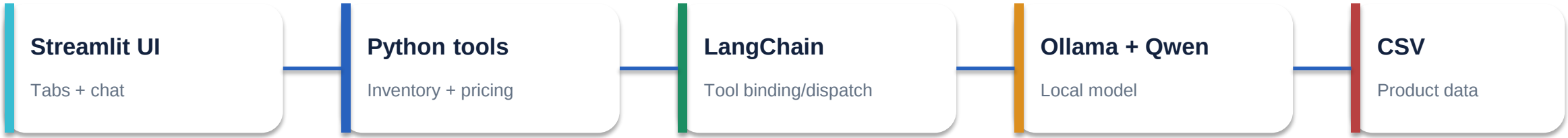
Technical architecture

Overview



Prototype architecture

Current implementation and data flow.



Deterministic guardrails

Minimum 20% margin • budget cap • 7-day demo duration

Local execution

Qwen runs through Ollama on the developer machine; no cloud model API required.

The prototype proposes promotions; it does not launch campaigns.

Agent tools and guardrails

Python owns calculations; the model interprets requests and explains tool results.

Inventory tool

Ranks products by stock to surface potential clearance priorities.

Safe discount tool

Matches competitor pricing where possible while enforcing the 20% margin floor.

Budgeted plan tool

Selects products in stock-priority order if the illustrative marketing estimate fits.

Calendar + segment tools

Shows weekends, configured Dasara/Diwali dates, and illustrative customer segments.

Constraint: the 5% marketing-cost rate is a demo assumption, not a measured campaign cost.

Prototype demonstration

Current sample data and observed app output.

10	4,270	₹50,000	₹42,000	₹8,000
Products	Units in stock	Demo budget	Estimated cost	Budget remaining

Current selected products (with ₹50,000 budget)

- Mouse — 800 units; 10% discount; ₹450 promotional price; 44.44% margin
- Laptop Stand — 600 units; 11.11% discount; ₹800 promotional price; 37.50% margin
- Estimated marketing cost: ₹42,000; remaining budget: ₹8,000

Results are based on the current illustrative CSV and 5% cost assumption.

Limitations and next steps

Clear boundaries make the demo more credible.

Current prototype

- Sample inventory/pricing data
- 5% estimated marketing-cost assumption
- Sample customer segments
- Manual Dasara/Diwali dates for 2026

Future enhancements

- Real campaign cost and sales history
- Real customer segmentation
- Regional holiday data integration
- Cannibalization and product relationships
- Promotion simulation and campaign execution integration

Next milestone

Validate recommendations against real sales outcomes before production use.

RETAIL AI

Smarter promotions.
Clearer constraints.

An explainable prototype for inventory-aware, margin-safe, budget-conscious retail promotion planning.

THANK YOU


Thank You!